

Types of Research

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Types of Research

- Science/Research is a very broad topic/activity (think of an archaeologist, a psychologist and an engineer)
- The purpose of this lecture is to understand this diversity
- But a natural previous question is what is Science/Research?
- There are many non-equivalent reasonable definitions

What is Science?

Demarcation Problem

In philosophy of science and epistemology, the **demarcation problem** is the question of how to distinguish between science and non-science. It also examines the boundaries between science, pseudoscience and other products of human activity, like art and literature.

It is an **open problem**.

What is Science?

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Disclaimer

Here, we are going to start with the most broad definition where Science/Research is any topic/activity in which one can make a claim about a topic of interest and defend it with a rational argumentation

Types of Research (w.r.t. enquiry mode)

- **Theoretical:**

- **Structured** approach
- Applied to an **abstraction** of the real world
- the purpose is to **prove** statements
- Which abstractions are worth to study? beauty? usefulness?
- the analysis/argumentation is based on **definitions** and **formal logic** (deduction)
- Premises are definitions, axioms and previously proved statements (theorems)
- More common in **Mathematics**
- Some people claim that this is not research

Types of Research (w.r.t. enquiry mode)

- **Quantitative:**

- **Structured** approach,
- the purpose is to **quantify** (measure, classify, predict) events that happen in the physical world
- the analysis/argumentation is typically based on data and **Statistics**.
 - Descriptive statistics
 - Inferential statistics
- More common in **Natural** Science.

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- **Qualitative:**

- **Flexible** approach,
- the purpose is to **describe/explain** in a meaningful/convincing way
- the analysis/argumentation is typically based on meaningful examples and a meaningful **narrative**.
- More common in **Social** Science.
- Some people claim that this is not research

Example

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- Understand the improvement of LLMs over the last 5 years:
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- Is there any critical advantage of quantum computing over traditional computing?: Theoretical
- Can Large Language Models be free of hallucinations?: Theoretical
- Understand the improvement of LLMs over the last 5 years: Quantitative
- Understand the propagation and amplification of fake news over social networks: Quantitative (if you want to understand the dynamics of virality) or Qualitative (if you want to understand the Psychological mechanisms of users)

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- Understand what does it mean for a woman to live in a violent relationship: Qualitative
- Understand the effects of shelter homes in a community: Quantitative

In the rest of the lecture we will focus on **Quantitative** Research

Types of Research (w.r.t. when we observe)

- **Longitudinal Research:** Data is collected for a group of individuals over prolonged periods of time
- **Cross-sectional Research:** Data is collected for a group of individuals without considering time

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- cross-sectional: may be *confounded* by inter-personal features (e.g. location)
 - longitudinal: may be *confounded* by temporal features (e.g. football world cup, age)

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- **Experimental**: we make an **intervention** in the independent variable to study its effect in the
 - **Requirement**: We **can** intervene on the independent variable
 - e.g: A new drug has its intended effect?
 - **Causal relation** can, in general, be proved

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 - **Causal relation** can, in general, be proved
- **Observational**: we do not make an intervention in the data (just observe)
 - **Limitation**: We **cannot** intervene on the independent variable
 - e.g: Smoking causes lung Cancer?
 - **Correlation** can, in general, be proved
 - Causal relation can, in general, only be **conjectured** and based on assumptions about the domain

The difficulties of Interventions

Interventions may be impossible

Example (Astrophysics)

Studying the effects of a naturally occurring astronomical event on a distant galaxy (e.g., a supernova on the chemical composition of planets)

Example (Economy)

Studyin the effects of recession in public debt

The difficulties of Interventions

Interventions may be unethical

Example

Studying the psychological effects of childhood trauma on personality

The difficulties of Interventions

Interventions may be complicated

Example

A group of Scientist in the UK thought that reducing the size of glasses in pubs from pint to a 0.33 liters would reduce obesity. For the experiment, they asked 1700 pubs if would make the change and only 13 accepted (and one of them cheated)

[Find the whole story here](#)

The Difficulties of Selecting Populations

Definition (External Validity)

Some times we cannot guarantee that the sample that we use for the experiment or the observation is representative of the total population

This problem may happen both in **experimental** and **observational** research

Experimental Research

The ideal form of Experimental Research is to run RCT's

Random Control Trials (RCT)

- Find a representative sample of the population under study
- Assign randomly subjects to the different groups
- Because of randomness, the different groups are statistically equivalent w.r.t. every known or unknown variable
- Manipulate the independent variable in each different group to determine its effect in the dependent variable

The simplest experiment is the so-called A/B test. Design of experiments is a very complex topic beyond the scope of this course

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Experimental Research may also suffer from external validity problem

Example

When physicians recruit patients to participate in an experiment, patients must explicitly accept and that may bias the sample.

This example illustrates the cost and complexity of longitudinal experiments

Example (Nutrition)

The effect of nutrition on obesity (the US experiment).

[Find details here](#)

- Obs. Research can be used to **identify associations** (i.e., **correlations**) between variables
- Which in turn can be used to **conjecture causality**
- But observations are very tricky and may be misleading

Observational Research may be Tricky

Example inspired in the anti-vax movement (70's)

The smallpox vaccine sometimes produces reactions which can be lethal. Suppose a country where out of 1 million children, 99 percent are vaccinated. Further, consider that 99 children die from the reaction and 40 from smallpox. Is vaccination effective?

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- In this case it makes sense to adjust values to the size of the population. Without any vaccination we should expect 3861 deads from smallpox. Probably many more, because those not vaccinated in the example, benefit from the vaccinated (they make the propagation much more unlikely).
- But may be the two populations (vaccinated vs non-vaccinated) are different w.r.t. other relevant variables.

Observational Research may be tricky

This is a true story that went viral during Covid

During the first months after covid vaccination

- **Data** (from a research paper): Considering covid patients at UK hospitals, 0.15% of non-vaccinated died compared with 1% of vaccinated people.
- **Headline in some media:** "vaccinated people is 6 times more likely to die of covid"
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The headline makes general claims from a very specific population

- 1 Vaccinated people are way less likely to get infected
- 2 Vaccination is much more frequent among old or vulnerable people

Observational Research may be tricky

Observation

In a sub-set of the population the correlation can be reversed

In the example, in the general Population, the correlation between vaccination and dying is negative. In the hospital population, the correlation between vaccination and dying is positive.

Observational Research can be Tricky

Basketball Shooting Performance

Consider two NBA basketball players. The performance of player A in the last three seasons is 25.3%, 32.1% and 32.9%. The performance of player B the same seasons is 25%, 31.4% and 29.1%. Who does it have the best performance?

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	Season 1	Season 2	Season 3	all 3 years
A	$104/411 = 25.3\%$	$45/140 = 32.1\%$	$163/495 = 32.9\%$	$312/1046 = 29.8\%$
B	$12/48 = 25\%$	$183/582 = 31.4\%$	$190/654 = 29.1\%$	$385/1284 = 30\%$

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Explanation:

$$\frac{a_1}{A_1} > \frac{b_1}{B_1} \wedge \frac{a_2}{A_2} > \frac{b_2}{B_2}$$

does not imply

$$\frac{a_1 + a_2}{A_1 + A_2} > \frac{b_1 + b_2}{B_1 + B_2}$$

Wrapping up about Experimental vs Observational Research

- Experimental Research is about manipulating Nature
- It is more powerful than Observational Research.
- But doing interventions may not be possible or reasonable
- Observational Research relies on what Nature let's you see
- Observations can be misleading

Types of Observational Research

Observational Research:

- **Descriptive**
- **Correlational**
- **Causal-comparative** (can prove, to a certain extent, causality)

- Considers **one variable** of interest
- the variable is **not manipulated**
- Seeks to **describe** its status
- Designed to provide **systematic information** about a phenomenon.
- The researcher does not usually begin with a hypothesis, but is likely to develop one after collecting the data.

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Examples:

- A description of how second-grade students spend their time during summer vacation
- A description of the tobacco use habits of teenagers
- A description of the attitudes of scientists regarding global warming

- Considers **two or more variables** of interest
- **Variables are not manipulated.**
- Determines the extent of their **relationship** using statistical data.
- Identifies **trends** and **patterns**
- Does not prove the cause of them
- Cannot be sure if they are **spurious** (i.e, not real, not what it appears to be)
- The scientist may use it to see correlation as a first step before understanding the connection.

Causal-comparative Research (Quasy or natural experiments)

- Considers two or more variables that are not manipulated and **Attempts to establish cause-effect relationships among the variables.**
- The researcher cannot randomly assign groups and must use ones that are **naturally** formed.
- The strenght of the experiment will depend on **how close to randomness** nature seemed to behave.
- **determining causality must be done carefully**, as other variables, both known and unknown, could still affect the outcome.

Natural Experiments: examples

Example

Cognitive consequences on children of mother's malnutrition during pregnancy.

Natural Experiments: examples

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Cognitive consequences on children of mother's malnutrition during pregnancy.

Study children of Muslim mothers whose pregnancy included Ramadan

Example

Effects of raising minimum wages on employment.

Natural Experiments: examples

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Study children of Muslim mothers whose pregnancy included Ramadan

Example

Effects of raising minimum wages on employment.

Study employment dynamics of two neighbor/similar regions after there is an increment of the minimum wage in one of them

[Find more examples here](#)

Types of Disciplines

- How Easy is to observe/intervene?
 - Computer Science
 - Biology
 - Economy
 - History

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- How Easy is to observe/intervene?
 - Computer Science
 - Biology
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- How Easy is to make good claims? (precise and falsifiable)
 - Computer Science
 - Biology
 - Environmental Sciences
 - Psychology
 - History

Types of Research (w.r.t. motivation)

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- **Basic Research:** Understanding the world for the sake of curiosity
- **Applied Research:** Discovering how to take advantage of our understanding of the world
- **Development and Innovation:** Making applied research readily available

The frontier is not clear.

Example: Electromagnetism

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When a new technology arises the bulk of research slowly moves from Basic to Innovation (we have seen it in Computer Science)

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- **Design Science** refers to a scientific (rational and systematic) **approach to designing solutions** (prototyping)
- **Design Science Research (DSR)** can be seen as the Scientific Method for building solutions (Applied Research)

Design Science Research (DSR)

Definition (According to the Design Science Journal)

Quantitative and **qualitative** research in the creation of **artifacts** and **systems**, and their embedding in our **physical, virtual, psychological, economic**, and **social** environment

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Example

- Physical: Design a robot to take care of elderly
- Virtual: Design an app to restrict internet access to children
- Psychological: Design a social network's interface to decrease the propagation of fake news
- Economic/Social: Design a policy to decrease rental price

Design Science Research (DSR)

- Fits very well into Engineering
- It was originated in 1957 (by R. Buckminster Fuller)
- Applied to Software and Information systems since 2010 (A. Hevner and S. Chatterjee)

- ➊ **Identification** of the problem and **justifying** the value of a solution;
- ➋ **Definition** of objectives for a solution;
- ➌ **Design** (typically modular design);
- ➍ **Development** of modules/components
- ➎ **Demonstration** by using the artifact to solve the problem;
- ➏ **Evaluation** of the solution, comparing the objectives and the actual observed results

It will only be **research** if it shows sufficient novelty

Identification of the problem

- Vehicles using combustion engines are an environmental problem
- Justify the potential value of replacing combustion engines in vehicles with data
 - Cars represent a ?? percentage of CO₂ emissions
 - If we eliminate pollution produced by cars, the consequences would be...

Definition of objectives

Our proposed solution is to design an Electric Vehicle such that:

- Range...
- Safety...
- Price ...
- Durability ...
- Environmental aspects ...
- ...

Example DSR

Design

Identify and **specify** different modules that need to be built (probably the most critical one is the battery)

Example DSR

Development

Implement every module

Demonstration and Evaluation

Testing modules in isolation

Assembling and testing prototype